

linear methods

The population of a community is known to increase at a rate proportional to the number of people present at time t . If an initial population P_0 has doubled in 6 years, how long (in years) will it take to triple? (Round your answer to one decimal place.)

the DE needed for this problem is:

$$\frac{dP}{dt} = kP$$

• you can solve this DE by separation of variables

$$\frac{1}{P} dP = k dt$$

$$\ln|P| = kt + C$$

$$e^{\ln|P|} = e^{kt+C}$$

$$P(t) = e^{kt} \cdot e^C$$

$$\text{let } e^C = P_0$$

When asked about population use this
No need to resolve the DE

$$P(t) = P_0 e^{kt}$$

initial Population

the problem tells us "if the initial population has doubled in 6 years how long will it take to triple"

$$P(6) = 2P_0 \quad \bullet \text{ Plug this into equation}$$

$$2P_0 = P_0 e^{6k}$$

$$2 = e^{6k}$$

$$\ln(2) = 6k \ln(e)$$

$$k = \frac{\ln(2)}{6}$$

• For linear problems they must give you at least one condition so you can find k . For this problem it was the initial population doubled in 6 years but the actual question is how long will it take to triple

"k" tells you the rate or pattern for how the quantity changes over time this "k" is true for all time stamps not just one

Now to solve the actual question: $P(6) = 3P_0$

$$3P_0 = P_0 e^{\left(\frac{\ln(2)}{6}\right)t}$$

• we are solving for t

$$3 = e^{\left(\frac{\ln(2)}{6}\right)t}$$

$$3 = (2^{1/6})^t$$

$$\ln(3) = t \ln(2^{1/6})$$

$$t = \frac{\ln(3)}{\ln(2^{1/6})} = 9.5 \text{ years}$$

The population of bacteria in a culture grows at a rate proportional to the number of bacteria present at time t . After 3 hours it is observed that 600 bacteria are present. After 10 hours 2000 bacteria are present. What was the initial number of bacteria? (Round your answer to the nearest integer.)

$$\frac{dP}{dt} = kP$$

$$\rightarrow P(t) = P_0 e^{kt}$$

$$P(3) = 600$$

$$P(10) = 2000$$

} you are given this and told find the initial number of bacteria

$$600 = P_0 e^{3k}$$

and

$$2000 = P_0 e^{10k}$$

isolate P_0 :

$$P_0 = 600 e^{-3k}$$

and

$$P_0 = 2000 e^{-10k}$$

Set them equal to each other to get "k"

$$600 e^{-3k} = 2000 e^{-10k}$$

$$\frac{e^{-3k}}{e^{-10k}} = \frac{10}{3}$$

$$e^{7k} = \frac{10}{3}$$

$$7k \ln(e) = \ln\left(\frac{10}{3}\right)$$

$$k = \frac{1}{7} \ln\left(\frac{10}{3}\right)$$

Plug this back into any of the two equations

$$P_0 = 600 e^{-3 \left(\frac{\ln(10/3)}{7} \right)}$$

$$P_0 = 365 \text{ bacteria}$$

The radioactive isotope of lead, Pb-209, decays at a rate proportional to the amount present at time t and has a half-life of 3.3 hours. If 1 gram of this isotope is present initially, how long (in hr) will it take for 75% of the lead to decay? (Round your answer to two decimal places.)

Notice the lead will decrease so you need to add a negative sign to the DE

$$\begin{aligned} \frac{dP}{dt} &= -kP \\ \rightarrow P(t) &= P_0 e^{-kt} \end{aligned}$$

$P(3.3) = 1/2$ • When half-life is mentioned all that means is $1/2$ the P_0 (initial amount) is 1 gram so, $P_0 = 1$

$$1/2 = e^{-3.3k}$$

$$\ln(1/2) = -3.3k \ln(e)$$

$$\ln(1) - \ln(2) = -3.3k$$

apply log rules

$$k = \frac{\ln(2)}{3.3}$$

you are told "how long will it take for 75% to decay" 75% decays you are left with 25%

$$P(t) = 25\%$$

$$P(t) = P_0 e^{-kt}$$

$$25 = e^{-\left(\frac{\ln(2)}{3.3}\right)t}$$

$$\ln(25) = -\frac{\ln(2)}{3.3} t \quad \ln(e)$$

$$-\frac{3.3 \ln(25)}{\ln(2)} = t$$

• time cannot be negative

$$t = 6.6 \text{ hours}$$

Initially 100 milligrams of a radioactive substance was present. After 6 hours the mass had decreased by 7%. If the rate of decay is proportional to the amount of the substance present at time t , find the amount (in milligrams) remaining after 24 hours. (Round your answer to one decimal place.)

$$\frac{dP}{dt} = -kP$$
$$\rightarrow P(t) = P_0 e^{-kt}$$

• there is a negative because it is decreasing

initially there is 100 mg of substance so:

$$P_0 = 100 \text{ mg}$$

decreased by 7% meaning there is 93% left

$$P(6) = 93 \rightarrow$$

$$93 = 100 e^{-6k}$$

$$0.93 = e^{-6k}$$

$$\ln(0.93) = -6k \quad \ln(e)$$

$$k = \frac{\ln(0.93)}{-6}$$

We now want to find the amount after 24 hours $P(24) = ?$

$$P(24) = 100 e^{-24 \left(\frac{-\ln(0.94)}{-6} \right)}$$

$$P(24) = 74.8 \text{ mg}$$

Initially 100 milligrams of a radioactive substance was present. After 5 hours the mass had decreased by 6%. If the rate of decay is proportional to the amount of the substance present at time t , determine the half-life (in hours) of the radioactive substance. (Round your answer to one decimal place.)

$$\frac{dP}{dt} = -kP$$

• negative because its decreasing

$$P(t) = P_0 e^{-kt}$$

You are told that initially you have 100mg so:

$$P_0 = 100 \text{ mg}$$

if the mass decreased by 6% you have 94% left so:

$$P(5) = 94$$

$$94 = 100 e^{-5k}$$

$$0.94 = e^{-5k}$$

$$\ln(0.94) = -5k \ln(e)$$

$$k = \frac{\ln(0.94)}{-5}$$

the half life is the time it takes for half the substance to decay

derivation of half-life equation

$$P(t_{1/2}) = \frac{1}{2} P_0 \rightarrow P(t) = P_0 e^{-kt}$$
$$\cancel{\frac{1}{2} P_0} = \cancel{P_0} e^{-k t_{1/2}}$$
$$\frac{1}{2} = e^{-k t_{1/2}}$$

$$\ln(\frac{1}{2}) = -k t_{1/2}$$

$$\cancel{\ln(1)} - \ln(2) = -k t_{1/2}$$
$$-\ln(2) = -k t_{1/2}$$

$$t = \frac{\ln(2)}{k}$$

We already found k so now all we have to do is plug into the half-life equation

$$t_{1/2} = \frac{\ln(2)}{0.012375080}$$

$$t_{1/2} = 56 \text{ hours}$$

Part #2

A thermometer is removed from a room where the temperature is 70°F and is taken outside, where the air temperature is 20°F . After one-half minute the thermometer reads 60°F . What is the reading of the thermometer (in $^\circ \text{F}$) at $t = 1 \text{ min}$? (Round your answer to two decimal places.)

When you are dealing with Newton's laws the DE is:

$$\frac{dT}{dt} = -k(T - T_s)$$

↘

$$T(t) = T_s + (T_0 - T_s)e^{-kt}$$

T_0 = initial temperature, T_s = surrounding temperature

$T_0 = 70^\circ$
 $T_s = 20^\circ$

• You are also told "after one-half minute the thermometer reads 60°F " so:

$$T(0.5) = 60$$

$$T(t) = T_s + (T_0 - T_s)e^{-kt}$$

$$60 = 20 + (70 - 20)e^{-0.5k}$$

$$40 = 50e^{-0.5k}$$

$$0.8 = e^{-0.5k}$$

$$\ln(0.8) = -0.5k \quad \ln(e)$$

$$k = \frac{\ln(0.8)}{-0.5}$$

We want to find what the reading on the temperature will be at $t = 1$

$$T(t) = T_s + (T_0 - T_s)e^{-kt}$$

$$T(1) = 20 + (50)e^{-\left(\frac{\ln(0.8)}{-0.5}\right)t} \quad (1)$$

$$T(1) = 50^\circ$$

Same question but how long (in minutes) will it take for the thermometer to reach 35°F ?

$$T(t) = T_s + (T_0 - T_s)e^{-kt}$$

$$35 = 20 + (50)e^{-\left(\frac{\ln(0.8)}{-0.5}\right)t}$$

$$15 = 50e^{-\left(\frac{\ln(0.8)}{-0.5}\right)t}$$

$$0.3 = e^{\cancel{\ln(0.8)}^{-1/0.5}t}$$

$$0.3 = (0.8)^{-2}t$$

$$\ln(0.3) = t \ln(0.8^{-2})$$

$$t = \frac{\ln(0.3)}{\ln(0.8^{-2})} = 2.70 \text{ min}$$

A thermometer is taken from an inside room to the outside, where the air temperature is 25°F . After 1 minute the thermometer reads 75°F , and after 5 minutes it reads 50°F . What is the initial temperature (in $^{\circ}\text{F}$) of the inside room? (Round your answer to two decimal places.)

$$\frac{dT}{dt} = -k(T - T_s)$$

$$\rightarrow T(t) = T_s + (T_0 - T_s)e^{-kt}$$

$T_s = 25^{\circ}$
 $T(1) = 75^{\circ}$ } • you are given these two and asked to find the initial temperature or T_0

$$75 = 25 + (T_0 - 25)e^{-k}$$

$$50 = (T_0 - 25)e^{-k}$$

$$e^{-k} = \frac{50}{(T_0 - 25)}$$

equation #1

$T(5) = 50 \rightarrow$

$$T(t) = T_s + (T_0 - T_s)e^{-kt}$$

$$50 = 25 + (T_0 - 25)e^{-5k}$$

$$25 = (T_0 - 25)e^{-5k}$$

equation #2

Plug the e^{-k} from equation #1 to equation #2

$$25 = (T_0 - 25) \left(\frac{50}{T_0 - 25} \right)^5$$

$$25 = (T_0 - 25) \frac{(50)^5}{(T_0 - 25)^5}$$

$$25 = \frac{(50)^5}{(T_0 - 25)^4}$$

$$(T_0 - 25)^4 = \frac{(50)^5}{25}$$

$$T_0 = \left(\sqrt[4]{\frac{50}{25}} + 25 \right) = 84.46^{\circ}$$

$$\left(\begin{aligned} \frac{dT}{dt} &= -\kappa (T - T_s) \\ \rightarrow T(t) &= T_s + (T_0 - T_s)e^{-\kappa t} \end{aligned} \right.$$

We are looking for the oven's temperature " T_s " (surrounding)

$$\left. \begin{aligned} T_0 &= 75 \\ T(1/2) &= 110^\circ \end{aligned} \right\} \begin{aligned} T(t) &= T_s + (T_0 - T_s)e^{-\kappa t} \\ 110 &= T_s + (75 - T_s)e^{-0.5\kappa} \end{aligned} \quad \text{equation \#1}$$

$$\left. \begin{aligned} T_0 &= 75 \\ T(1) &= 140 \end{aligned} \right\} \begin{aligned} T(t) &= T_s + (T_0 - T_s)e^{-\kappa t} \\ 140 &= T_s + (75 - T_s)e^{-\kappa} \end{aligned} \quad \text{equation \#2}$$

equation #1

$$\begin{aligned} 110 &= T_s + (75 - T_s)e^{-0.5\kappa} \\ \ln(110 - T_s) &= \ln(75 - T_s) - 0.5\kappa \end{aligned}$$

equation #2

$$\begin{aligned} 140 &= T_s + (75 - T_s)e^{-\kappa} \\ \ln(140 - T_s) &= \ln(75 - T_s) - \kappa \end{aligned}$$

Now subtract equation #1 from equation #2

$$\begin{aligned} \ln(110 - T_s) &= \ln(75 - T_s) - 0.5\kappa \\ - \ln(140 - T_s) &= \ln(75 - T_s) - \kappa \\ \hline \ln(110 - T_s) - \ln(140 - T_s) &= -0.5\kappa + \kappa \end{aligned}$$

$$\ln \left(\frac{110 - T_s}{140 - T_s} \right) = -0.5 K$$

$$K = -2 \ln \left(\frac{110 - T_s}{140 - T_s} \right)$$

you can plug K back into any one of the equations

$$\ln(110 - T_s) = \ln(75 - T_s) - 0.5 \left(-2 \ln \left(\frac{110 - T_s}{140 - T_s} \right) \right)$$

$$\ln(110 - T_s) = \ln(75 - T_s) + \ln \left(\frac{110 - T_s}{140 - T_s} \right)$$

$$\cancel{\ln}(110 - T_s) = \cancel{\ln} \left(\frac{(75 - T_s)(110 - T_s)}{(140 - T_s)} \right)$$

• apply log laws on right side

$$110 - T_s = \frac{(75 - T_s)(110 - T_s)}{(140 - T_s)}$$

$$(110 - T_s)^2 = (75 - T_s)(140 - T_s)$$

$$T_s^2 - 220T_s + 12100 = T_s^2 - 215T_s + 10500$$

$$5T_s = 1600$$

$$\boxed{T_s = 320^\circ}$$

A tank contains 300 liters of fluid in which 50 grams of salt is dissolved. Pure water is then pumped into the tank at a rate of 5 L/min; the well-mixed solution is pumped out at the same rate. Find the number $A(t)$ of grams of salt in the tank at time t .

for this problem we will use:

$$\frac{dA}{dt} = (\text{rate in}) - (\text{rate out})$$

rate in: **entering salt**

(flow in) (concentration in)

$$(0) (0) = 0$$

rate out:

(concentration in tank) (salt leaving per min)

$$\left(\frac{A}{300}\right) (5) = \frac{A}{60}$$

$$\frac{dA}{dt} = \left(0 - \frac{A}{60}\right) \Rightarrow \frac{dA}{dt} = -\frac{A}{60}$$

• solve this DE by
separation of variables

$$\int \frac{1}{A} dA = \int -\frac{1}{60} dt$$

$$\ln(A) = -\frac{t}{60} + c$$

$$e^{\ln(A)} = e^{-\frac{t}{60} + c}$$

$$A = e^{-t/60} e^c$$

$$\text{let } e^c = C_1$$

$$A(t) = C_1 e^{-t/60}$$

$$A(0) = 50$$

• if you aren't
give a time
assume 0

$$50 = C_1 e^{-0/60} \rightarrow C_1 = 50$$

$$A(t) = 50 e^{-t/60}$$

A large tank is filled to capacity with 700 gallons of pure water. Brine containing 3 pounds of salt per gallon is pumped into the tank at a rate of 7 gal/min. The well-mixed solution is pumped out at the same rate. Find the number $A(t)$ of pounds of salt in the tank at time t .

$$\frac{dA}{dt} = R_{in} - R_{out}$$

$$R_{in} = \left(\begin{array}{c} \text{Concentration} \\ \text{of salt in} \\ \text{Flow} \end{array} \right) \cdot \left(\begin{array}{c} \text{input rate} \\ \text{of brine} \end{array} \right) \Rightarrow (3) (7) = 21$$

= input rate of salt

$$R_{out} = \left(\begin{array}{c} \text{Concentration} \\ \text{of salt out} \\ \text{Flow} \end{array} \right) \left(\begin{array}{c} \text{output rate} \\ \text{of brine} \end{array} \right) \Rightarrow \left(\frac{A}{100} \right) (7) = \frac{A}{100}$$

= output rate of salt

you must consider
what is in the
tank

$$\frac{dA}{dt} = 21 - \frac{A}{100}$$

• this will be solved by
linear equations

$$\frac{dA}{dt} + \frac{A}{100} = 21$$

• this is now in standard
form

$$P(t) = \frac{1}{100}, \quad e^{\int 1/100 dt} \Rightarrow e^{t/100}$$

$$e^{t/100} \cdot \frac{dA}{dt} + e^{t/100} \cdot \frac{A}{100} = 21 e^{t/100}$$

$$\int \frac{d}{dt} \left[e^{t/100} A \right] = \int 21 e^{t/100}$$

$$u = t/100$$

$$du = \frac{1}{100} dt$$

$$100 du = dt$$

$$e^{t/100} A = 2100 e^{t/100} + C$$

$$A(t) = 2100 + C e^{-t/100}$$

• the question says a large tank is filled to capacity of 700 gallons of pure water so there is no salt in the tank so:

$$A(0) = 0$$

$$A(t) = 2100 + C e^{-t/100}$$

$$0 = 2100 + C e^0$$

$$C = -2100 \quad \bullet \text{ Plug this back in to equation}$$

$$A(t) = 2100 - 2100 e^{-t/100}$$

how to solve the newton heating and cooling DE

$$\frac{dT}{dt} = -\kappa (T - T_s)$$

$$\frac{1}{(T - T_s)} dT = -\kappa dt$$

$$\int \frac{1}{(T - T_s)} dT = \int -\kappa dt$$

$$u = T - T_s \quad = -\kappa t + C$$
$$du = dT$$

$$\int \frac{1}{u} dT = -\kappa t + C$$

$$\ln |u| = -\kappa t + C$$

$$\ln |T - T_s| = -\kappa t + C$$

$$\cancel{e^{\ln}} |T - T_s| = e^{-\kappa t + C}$$

$$T - T_s = e^{-\kappa t} \cdot e^C$$

$$T(t) = T_s + C_1 e^{-\kappa t} \quad \text{where } C_1 = (T_0 - T_s)$$

$$T(t) = T_s + (T_0 - T_s) e^{-\kappa t}$$

- if the temperature is decreasing you need a negative in front of κ
if the temperature is increasing you don't need a negative

$$\text{let } e^C = C$$